

Advanced (Habanero)

- Q1.** What is the graph of the system $x + y \geq 2$ and $x - y \leq 1$?
- (A) Two intersecting lines with shaded regions
 (B) Two parallel lines with no shaded regions
 (C) A single line with a shaded region
 (D) No lines with a shaded region
 (E) A circle with a shaded region
- Q2.** Which of the following is the solution region for the system $x \geq 1$ and $y \leq 2$?
- (A) Upper left quadrant
 (B) Upper right quadrant
 (C) Lower left quadrant
 (D) Lower right quadrant
 (E) Entire coordinate plane
- Q3.** If the system $x + y \leq 3$ and $x - y \geq -1$ has a solution, what is the relationship between x and y ?
- (A) $x \geq y$
 (B) $x \leq y$
 (C) $x = y$
 (D) $x \neq y$
 (E) x and y are unrelated
- Q4.** What is the equation of the boundary line for the inequality $2x - 3y \geq 5$?
- (A) $2x - 3y = 5$
 (B) $2x + 3y = 5$
 (C) $x - y = 1$
 (D) $x + y = 2$
 (E) $y = 2x - 1$
- Q5.** The system of inequalities $x \geq 0$, $y \geq 0$, and $x + y \leq 4$ represents a
- (A) Triangle
 (B) Rectangle
 (C) Circle
 (D) Trajectory
 (E) Parabola
- Q6.** For the system $y \geq x - 2$ and $y \leq -x + 1$, the solution region is
- (A) A triangle with vertices at $(0, -2)$, $(1, -1)$, and $(0, 1)$
 (B) A triangle with vertices at $(0, 0)$, $(1, 1)$, and $(-1, -1)$
 (C) A rectangle with vertices at $(0, 0)$, $(2, 0)$, $(2, 1)$, and $(0, 1)$
 (D) A circle centered at the origin with radius 2
 (E) The entire coordinate plane
- Q7.** What is the solution to the system $x + 2y \geq 4$ and $x - 2y \leq -2$?
- (A) The point $(2, 1)$
 (B) The line $x = 2$
 (C) The line $y = 1$
 (D) The region bounded by the lines $x = 2$ and $y = 1$
 (E) No solution
- Q8.** For the system of inequalities $y \geq x^2 - 4$ and $y \leq -x^2 + 4$, the solution region is
- (A) The region between the two parabolas
 (B) The region above the parabola $y = x^2 - 4$
 (C) The region below the parabola $y = -x^2 + 4$
 (D) The two parabolas intersect at a single point
 (E) There is no solution region

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + 2y \geq 3$ and $x - y \leq 1$. Identify the vertices of the solution region.
- Q10.** Find the solution region for the system $y \geq x - 2$ and $y \leq -x + 3$. Determine the area of the solution region.

Chef's Tips

- Tip 1: Ensure students understand how to graph multiple inequalities on the same coordinate plane.
- Tip 2: Emphasize the importance of identifying overlapping solution regions.
- Tip 3: Encourage students to interpret feasible regions in the context of real-world problems.